

import libraries

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [10]: filepath = 'https://raw.githubusercontent.com/datasciencedojo/datasets/master/passengers.csv'
```

```
In [11]: df = pd.read_csv(filepath)
```

```
In [13]: df.columns
```

```
Out[13]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
               'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
              dtype='object')
```

```
In [163... # To convert the dataframe into csv file.
path = 'D:\Python For data Analysis - IBM\To upload on linkedin\Day1-22nd Sept-Data wrangling\passenger.csv'
df.to_csv(path)
```

```
In [164... df.head(5)
```

```
Out[164...
```

	PassengerId	Survived	Pclass	Passenger_Name	Sex	Age	SibSp	Parch	Fare	Embarked	AgeGroup
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	7.2500	S	Young Adult
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	71.2833	C	Adult
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	7.9250	S	Young Adult
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	53.1000	S	Adult
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	8.0500	S	Adult

```
In [165... # Basic info: rows, columns, data types, missing values summary  
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 891 entries, 0 to 890  
Data columns (total 11 columns):  
#   Column          Non-Null Count  Dtype  
---  -  
0   PassengerId      891 non-null    int64  
1   Survived         891 non-null    int64  
2   Pclass          891 non-null    int64  
3   Passenger_Name   891 non-null    object  
4   Sex              891 non-null    category  
5   Age              891 non-null    float64  
6   SibSp            891 non-null    int64  
7   Parch           891 non-null    int64  
8   Fare             891 non-null    float64  
9   Embarked         889 non-null    category  
10  AgeGroup         817 non-null    category  
dtypes: category(3), float64(2), int64(5), object(1)  
memory usage: 58.8+ KB  
None
```

```
In [166... # From above summary we can observed that there are three columns who has missing values. ['Age','Cabin',Embarked']
```

```
In [167... #There are two methods to detect mssing data.  
# df.isnull() #This will give us the missing values existance inside a column in the form boolean values.  
# df.notnull() #This will give us the not missing values existance inside a column in the form boolean values.
```

```
In [51]: # We can seperately calculate mis
```

```
In [50]: # There are total 714 missing values in Age column. We will replace these values by mean.
```

```
In [56]: missing_data = df.isnull()  
missing_data.head(5)
```

Out[56]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	False	False	False	False	False	False	False	False	False	False	True	False
1	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	True	False
3	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	True	False

```
In [59]: for column in missing_data.columns:
          print(f"Total missing values in :{column}:{missing_data[column].value_counts()}")
          print(" ")
```

Total missing values in :PassengerId:PassengerId
False 891
Name: count, dtype: int64

Total missing values in :Survived:Survived
False 891
Name: count, dtype: int64

Total missing values in :Pclass:Pclass
False 891
Name: count, dtype: int64

Total missing values in :Name:Name
False 891
Name: count, dtype: int64

Total missing values in :Sex:Sex
False 891
Name: count, dtype: int64

Total missing values in :Age:Age
False 714
True 177
Name: count, dtype: int64

Total missing values in :SibSp:SibSp
False 891
Name: count, dtype: int64

Total missing values in :Parch:Parch
False 891
Name: count, dtype: int64

Total missing values in :Ticket:Ticket
False 891
Name: count, dtype: int64

Total missing values in :Fare:Fare
False 891
Name: count, dtype: int64

Total missing values in :Cabin:Cabin

```
True      687
False     204
Name: count, dtype: int64
```

```
Total missing values in :Embarked:Embarked
False     889
True        2
Name: count, dtype: int64
```

```
In [62]: # Now we will replace missing values in 'Age' column. From above summary we have found that there are total 177 missing values in 'Age' column.
# We will replace these missing values by mean.
```

```
In [82]: avg_age = round(df['Age'].astype('float').mean(axis=0),2) # astype() is important when the column data type is object
print("Average of 'Age' Column in",avg_age)
```

```
Average of 'Age' Column in 29.7
```

```
In [83]: df['Age'].replace(np.nan,avg_age,inplace=True)
```

```
In [84]: df['Age'].isnull().value_counts()
```

```
Out[84]: Age
False     891
Name: count, dtype: int64
```

```
In [85]: # Now there are no missing values in 'Age' Column. We can now check unique values in this column.
df['Age'].unique()
```

```
Out[85]: array([22.      , 38.      , 26.      , 35.      , 29.69911765,
 54.      , 2.      , 27.      , 14.      , 4.      ,
 58.      , 20.     , 39.      , 55.      , 31.      ,
 34.      , 15.      , 28.      , 8.       , 19.      ,
 40.      , 66.      , 42.      , 21.      , 18.      ,
 3.       , 7.       , 49.      , 29.      , 65.      ,
 28.5     , 5.       , 11.      , 45.      , 17.      ,
 32.      , 16.      , 25.      , 0.83     , 30.      ,
 33.      , 23.      , 24.      , 46.      , 59.      ,
 71.      , 37.      , 47.      , 14.5     , 70.5     ,
 32.5     , 12.      , 9.       , 36.5     , 51.      ,
 55.5     , 40.5     , 44.      , 1.       , 61.      ,
 56.      , 50.      , 36.      , 45.5     , 20.5     ,
 62.      , 41.      , 52.      , 63.      , 23.5     ,
 0.92     , 43.      , 60.      , 10.      , 64.      ,
 13.      , 48.      , 0.75     , 53.      , 57.      ,
 80.      , 70.      , 24.5     , 6.       , 0.67     ,
 30.5     , 0.42     , 34.5     , 74.      ,  ])
```

```
In [89]: # We do not need 'Ticket' and 'Cabin' column for our data analysis.
# We will drop these less useful columns
```

```
In [90]: df.drop(columns = ['Ticket','Cabin'],inplace= True)
```

```
In [93]: df.columns
```

```
Out[93]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
      'Parch', 'Fare', 'Embarked'],
      dtype='object')
```

```
In [109... # Below columns have text values, like ['male','female'] and ['C','S','Q'].If they are object, it means they are stor
df[['Embarked','Sex']]
```

Out[109...

	Embarked	Sex
0	S	male
1	C	female
2	S	female
3	S	female
4	S	male
...	...	...
886	S	male
887	S	female
888	S	female
889	C	male
890	Q	male

891 rows × 2 columns

```
In [110... # We can convert above columns values into catergorical values for memory optimization and better performance during  
# Much less memory used (can be 50-80% less)  
# Faster groupby or value_counts operations
```

```
In [111... df['Sex'] = df['Sex'].astype('category')  
df['Embarked'] = df['Embarked'].astype('category')
```

```
In [112... print(df.dtypes)
```

```
PassengerId      int64
Survived          int64
Pclass           int64
Name             object
Sex              category
Age             float64
SibSp            int64
Parch            int64
Fare             float64
Embarked         category
dtype: object
```

```
In [117... # we will check if there are any duplicates present .
# Check which rows are duplicates (all columns)
duplicates = df.duplicated()
duplicates
```

```
Out[117... 0      False
1      False
2      False
3      False
4      False
...
886    False
887    False
888    False
889    False
890    False
Length: 891, dtype: bool
```

```
In [118... df
```

Out[118...

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.000000	1	0	7.2500	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.000000	1	0	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.000000	0	0	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.000000	1	0	53.1000	S
4	5	0	3	Allen, Mr. William Henry	male	35.000000	0	0	8.0500	S
...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.000000	0	0	13.0000	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.000000	0	0	30.0000	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	29.699118	1	2	23.4500	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.000000	0	0	30.0000	C
890	891	0	3	Dooley, Mr. Patrick	male	32.000000	0	0	7.7500	Q

891 rows × 10 columns

In [119...

```
df.rename(columns = {'Name':'Passenger_Name'},inplace = True)
# Now check the columns
df.columns
```

Out[119...

```
Index(['PassengerId', 'Survived', 'Pclass', 'Passenger_Name', 'Sex', 'Age',
      'SibSp', 'Parch', 'Fare', 'Embarked'],
      dtype='object')
```

In [154...

```
# Define bins and labels
bins = [0,18,30,50]
labels = ['Child','Young Adult','Adult']
```

In [155...

```
df['AgeGroup'] = pd.cut(df['Age'], bins=bins, labels=labels, right=False)
```

```
In [156... df.head(5)
```

Out[156...

	PassengerId	Survived	Pclass	Passenger_Name	Sex	Age	SibSp	Parch	Fare	Embarked	AgeGroup
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	7.2500	S	Young Adult
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	71.2833	C	Adult
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	7.9250	S	Young Adult
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	53.1000	S	Adult
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	8.0500	S	Adult

In [159... df

Out[159...

	PassengerId	Survived	Pclass	Passenger_Name	Sex	Age	SibSp	Parch	Fare	Embarked	AgeGroup
0	1	0	3	Braund, Mr. Owen Harris	male	22.000000	1	0	7.2500	S	Young Adult
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.000000	1	0	71.2833	C	Adult
2	3	1	3	Heikkinen, Miss. Laina	female	26.000000	0	0	7.9250	S	Young Adult
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.000000	1	0	53.1000	S	Adult
4	5	0	3	Allen, Mr. William Henry	male	35.000000	0	0	8.0500	S	Adult
...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.000000	0	0	13.0000	S	Young Adult
887	888	1	1	Graham, Miss. Margaret Edith	female	19.000000	0	0	30.0000	S	Young Adult
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	29.699118	1	2	23.4500	S	Young Adult
889	890	1	1	Behr, Mr. Karl Howell	male	26.000000	0	0	30.0000	C	Young Adult
890	891	0	3	Dooley, Mr. Patrick	male	32.000000	0	0	7.7500	Q	Adult

891 rows × 11 columns

In [162...

```
df.to_csv('D:\Python For data Analysis - IBM\To upload on linkedin\Day1-22nd Sept-Data wrangling\\passengers_clean.csv')
```

Finally we have cleaned csv file, without missing values

In []: